PROJECT PROPOSAL

AI-Powered Research & Innovation Copilot

"Search Less. Solve More. with iNSIGHTS Layer 2"

TEAM NAME & ID

Innovation Systems Architecture Group
(ISAG-404)

INSTITUTION

Advanced College of Computational Sciences and Engineering

Problem & Existing Gaps

- **The Core Problem:** Students face a severe cognitive bottleneck transitioning from abstract academic ideation to practical engineering execution.
- **Who is Affected:** CS/SE students globally, highly susceptible to project abandonment due to compounded socio-economic and cognitive stressors.
- **Current Solutions:** Heavy reliance on generic search engines, disjointed static documentation, and manual tracking spreadsheets.
- **Key Limitations:** Extreme information asymmetry, hallucinated/deprecated code references, and a complete lack of context-aware, real-time mentorship.

The Ideation-to-Execution Bottleneck

Ideation & Theory
Literature review, abstract logic

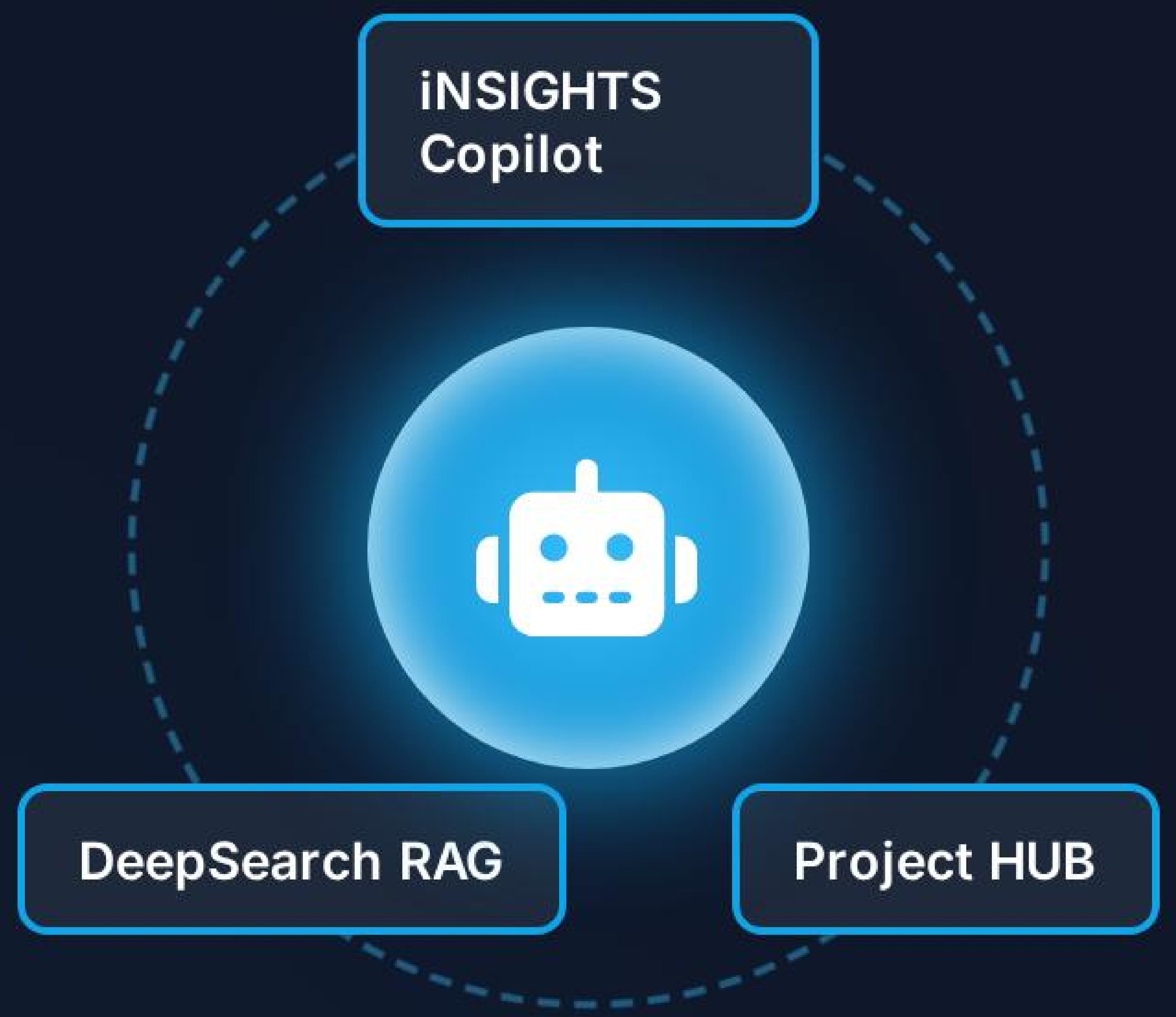
✖

Cognitive Overload
Information asymmetry, deprecated docs, stress

✖

Practical Execution
Deployment, debugging, delivery (Stalled)

Proposed Solution



A centralized intelligence core bridging the gap between human ideation and technical deployment.

- **The Core Idea:** An autonomous digital technical lead and project manager natively operating on the iNSIGHTS Layer 2 framework.
- **How It Solves the Problem:** Automatically translates natural language problem statements into comprehensive, implementation-ready architectural blueprints and structured agile sprints.
- **The Competitive Edge:** Unlike generic LLMs, it utilizes verifiable **Retrieval-Augmented Generation (RAG)** to eliminate hallucinations, ensuring code architectures are modern, secure, and logically sound.
- **Continuous Context:** Maintains persistent awareness of the student's project state, acting as a tireless mentor.

Architecture & Tech Stack

Frontend UI
React.js, Next.js, Tailwind
User Dashboards

→

API Gateway
Node.js, Express, GraphQL
Routing & Auth

→

AI Processing (L2)
Python, FastAPI, Llama-3
LangChain & DeepSearch

Pinecone
Vector Semantics

PostgreSQL
User State & ACID

Key Innovative Features

Multimodal DeepSearch

Executes semantic searches across text, code, and statistical databases. Synthesizes unstructured data into coherent, citation-backed knowledge clusters to validate ideas before coding begins.

Automated Project HUB

Dynamically generates complete SDLC blueprints, resolves software dependencies automatically, and outputs time-bound milestone plans tailored specifically to the user's declared skill level.

Collaborative AI Agents

Deploys continuous, personalized scrum masters natively via Telegram/Discord. Provides real-time, context-aware debugging assistance without requiring the student to switch contexts.

Real-time Web Intelligence

Continuously monitors repositories and package managers. Proactively alerts users if chosen tech stacks become deprecated or face security vulnerabilities, suggesting modern alternatives.

Impact & Feasibility

- **Beneficiaries:** CS/SE students globally, acting as a massive equalizer for underfunded educational institutions lacking continuous faculty mentorship.
- **Real-World Impact:** Standardizes the ideation-to-execution pipeline, fundamentally increasing project completion rates and fostering engineering confidence.
- **Scalability:** Designed for extreme scalability through stateless microservices and auto-scaling cloud infrastructure.
- **Cost Feasibility:** Achieves near-zero idle costs by utilizing serverless vector databases (Pinecone) and routing simpler queries to quantized open-source language models.

↓ 40%
CYCLE TIME REDUCTION

↑ 2x
PROJECT COMPLETION

Serverless Deployment
HIGH AVAILABILITY, LOW COST

Prototype Output & Conclusion

Why Select iNSIGHTS?

The Copilot bridges the academic chasm by turning vague ideas into structured engineering sprints instantly. It proves its value by tackling real-world complexities—like predicting hostel food waste—through automated, verifiable AI orchestration.

> ***"A comprehensive paradigm shift that guarantees students can finally Search Less and Solve More."***